



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH
(AIUB)

COMPUTER NETWORKS

COMPUTER NETWORKS LAB ASSIGNMENT

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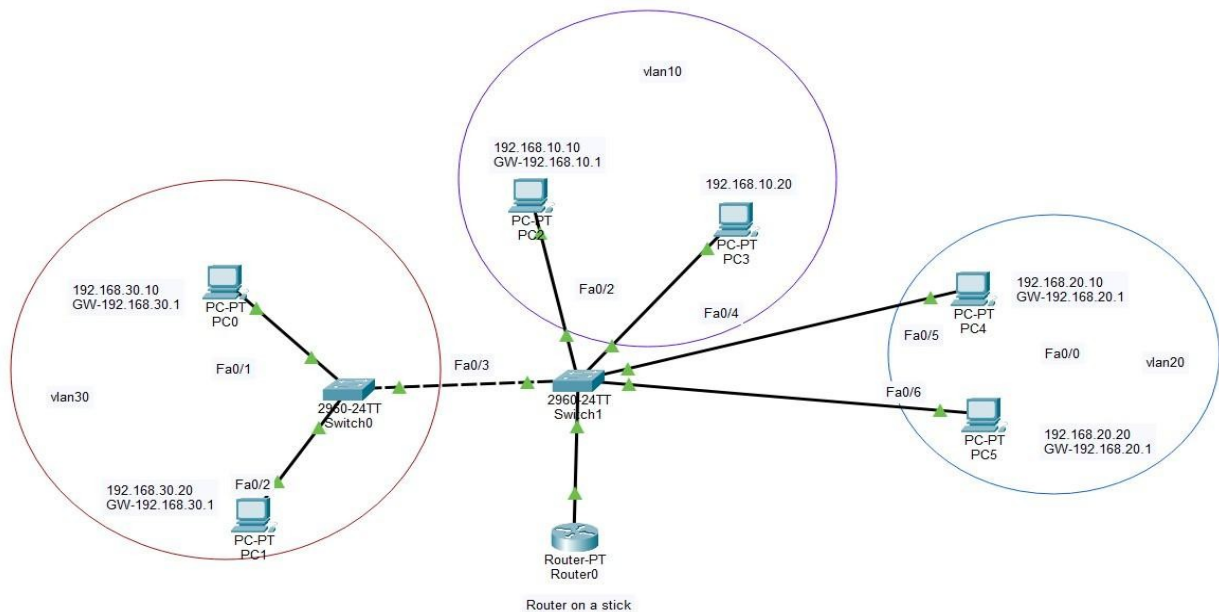
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1. Network Segmentation Using VLANs and Inter-VLAN Routing via Router-on-a-Stick



Configuration:

A segmented network was constructed to demonstrate inter-VLAN routing using the Router-on-a-Stick method. The physical topology consisted of six end-user PCs, two Cisco 2960-24TT layer 2 switches, and a single router.

The logical design established three distinct Virtual LANs (VLANs), each mapped to its own IP subnet:

VLAN 10 was assigned the 192.168.10.0/24 network.

VLAN 20 was assigned the 192.168.20.0/24 network.

VLAN 30 was assigned the 192.168.30.0/24 network.

Switch and End-Device Configuration:

On the first switch (Switch0), two PCs were connected. PC0 (192.168.30.10) was placed on port FastEthernet0/1 and PC1 (192.168.30.20) on port FastEthernet0/2. Both interfaces were configured as access ports and assigned to VLAN 30, with their default gateway set to 192.168.30.1.

The second switch (Switch1) hosted the remaining devices across two VLANs. For VLAN 10, PC2 (192.168.10.10) and PC3 (192.168.10.20) were connected to access ports FastEthernet0/2 and FastEthernet0/4, respectively. For VLAN 20, PC4 (192.168.20.10) and PC5 (192.168.20.20) were connected to access ports FastEthernet0/5 and FastEthernet0/6. The PCs in each VLAN were configured with the corresponding default gateway (192.168.10.1 for VLAN 10, 192.168.20.1 for VLAN 20).

Trunking and Router-on-a-Stick Setup:

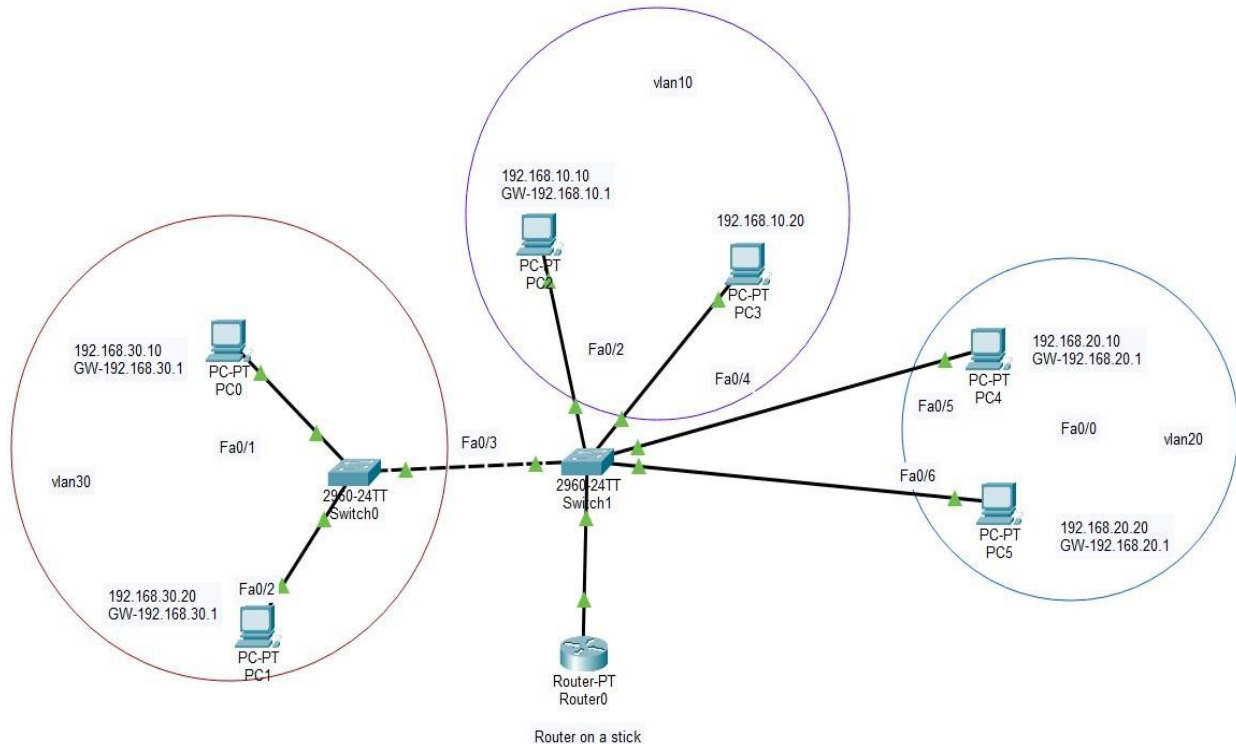
To interconnect the switches, a copper crossover cable established a link between them, with both ends configured as trunk ports to carry traffic for all VLANs. Switch1 was then connected to the router, with its uplink port also set to trunk mode.

The core of the inter-VLAN routing was implemented on the router using the Router-on-a-Stick technique. A single physical interface on the router was divided into logical sub-interfaces, one for each VLAN (10, 20, and 30). Each sub-interface was assigned an IP address from its respective VLAN's subnet, specifically the .1 address, to serve as the default gateway for devices in that VLAN.

Verification and Testing

Following the configuration, connectivity tests were performed. The results confirmed successful communication between devices residing within the same VLAN. Furthermore, devices located in different VLANs (for example, a PC in VLAN 10 communicating with a PC in VLAN 20) were able to exchange data seamlessly, demonstrating that the inter-VLAN routing through the router was functioning correctly.

2. Design and Implementation of a Segmented Multi-Switch VLAN Network:



Configuration:

A practical network was set up to segment traffic across three distinct VLANs using a multi-switch architecture. The physical hardware included six end-user computers, two layer-2 switches (Cisco 2960-24TT models), and a single router for inter-VLAN routing.

The logical segmentation divided the network into the following groups:

VLAN 10 used the 192.168.10.0/24 subnet.

VLAN 20 used the 192.168.20.0/24 subnet.

VLAN 30 used the 192.168.30.0/24 subnet.

The endpoints were distributed across the switches. On the first switch (Switch0), two PCs were allocated to VLAN 30. PC0, with IP address 192.168.30.10, was linked to port FastEthernet0/1, while PC1 (192.168.30.20) was linked to port FastEthernet0/2. Both devices were configured to use 192.168.30.1 as their network gateway.

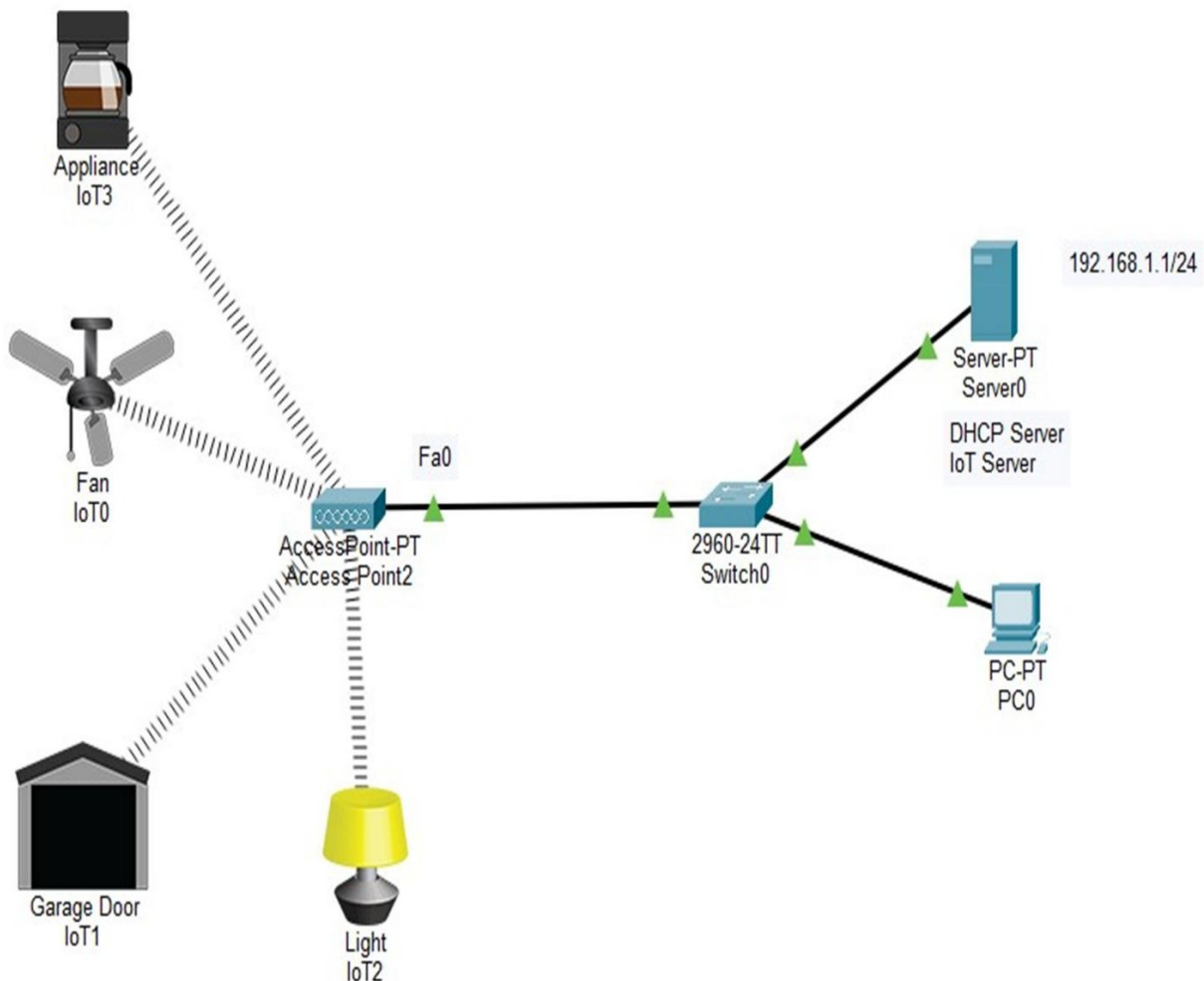
The second switch (Switch1) hosted devices for the remaining two VLANs. For VLAN 10, PC2 (192.168.10.10) and PC3 (192.168.10.20) were connected to ports FastEthernet0/2 and FastEthernet0/4, respectively, with their gateway set to 192.168.10.1. Similarly, for VLAN 20, PC4 (192.168.20.10) and PC5 (192.168.20.20) were attached to ports FastEthernet0/5 and FastEthernet0/6, using 192.168.20.1 as their gateway.

To extend the VLANs across the physical switches, a trunk link was established between Switch0 and Switch1 using a copper crossover cable. This trunk port was configured to carry traffic for all user VLANs. Switch1 was then connected to the router, with its uplink port also set to trunk mode.

The router was configured for the "Router-on-a-Stick" method to enable communication between the VLANs. This was achieved by creating three logical sub-interfaces on a single physical router port, one dedicated to each VLAN (10, 20, and 30). An IP address from the corresponding subnet was applied to each sub-interface, designating it as the default gateway for that specific VLAN. All switch ports directly connected to the end-user PCs were set to access mode and bound to their respective VLANs.

Upon completion, the network was tested. Connectivity checks confirmed that devices within the same VLAN could communicate directly. Furthermore, successful pings between devices in different VLANs (e.g., from a host in VLAN 10 to a host in VLAN 20) validated that the Router-on-a-Stick configuration was correctly routing traffic between the segmented network.

3. IoT Network Integration with Centralized Management



Configuration:

A practical network was implemented to support Internet of Things (IoT) devices, focusing on integration and centralized control. The topology connected multiple smart devices wirelessly to an access point, which was then integrated with a wired infrastructure consisting of a switch, a dedicated server, and a management PC.

The wireless segment included four distinct IoT devices: a smart appliance, a garage door opener, a fan, and a lighting system. These devices were configured to associate with a dedicated wireless access point, establishing their primary connection to the network.

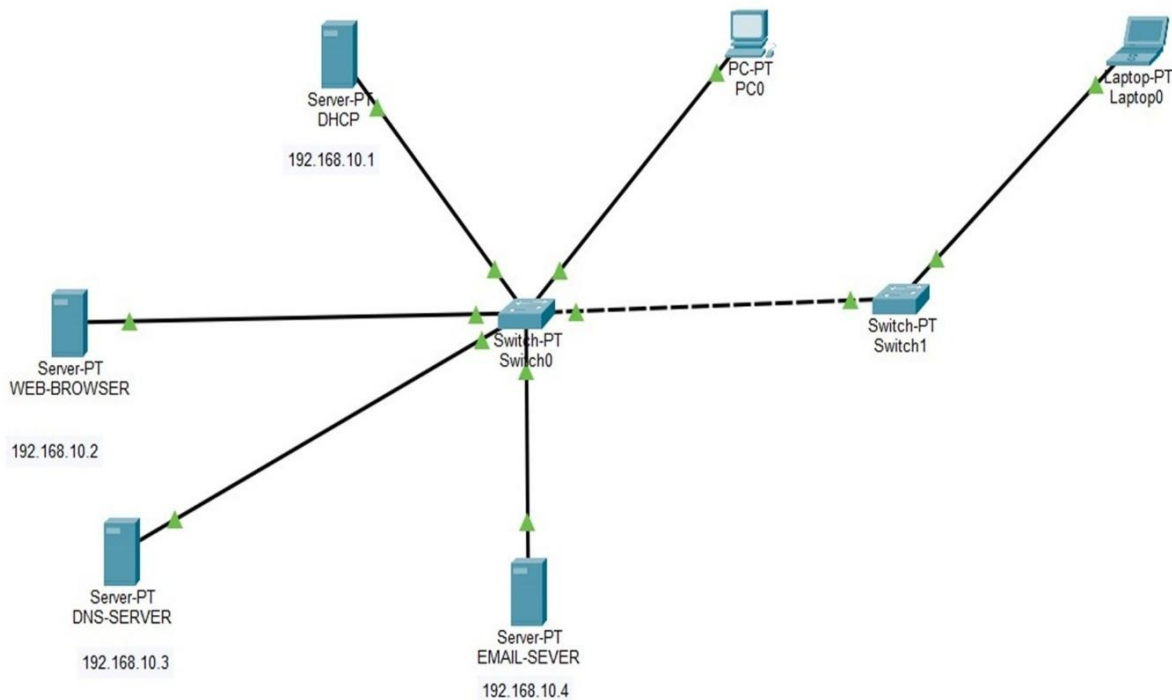
To bridge the wireless and wired domains, the access point was physically connected to a central switch (Switch1) using an Ethernet cable. This switch also hosted the network's management and server resources. A user workstation (PC0) was connected to the switch, providing an interface for monitoring and issuing commands to the IoT ecosystem. A server (Server0) was also attached to the switch and configured as a central hub for aggregating data from the IoT devices, running management

applications, and storing historical logs.

The access point was carefully configured to provide stable and secure wireless coverage for all IoT devices. Simultaneously, the layer 2 switch was set up to efficiently handle data traffic between the wired server, the management PC, and the uplink to the wireless access point, forming a cohesive network backbone.

After configuration, end-to-end functionality was validated. Testing confirmed that all IoT devices could successfully communicate with the central server. Furthermore, the management PC could reliably monitor device status and send control commands through the server, demonstrating a fully operational IoT monitoring and control system

4. Implementation of an Integrated Services Network with DHCP, DNS, Web, and Email



Configuration:

A functional network was built to demonstrate the integration of core network services. The infrastructure consisted of two interconnected layer 2 switches, a desktop PC, a laptop, and four dedicated servers, each hosting a critical service.

The physical network was established by connecting Switch0 and Switch1 with a copper crossover cable, creating the core backbone. To Switch0, four servers were connected via Ethernet and statically assigned IP addresses on the 192.168.10.0/24 subnet: a DHCP server (192.168.10.1), a web server (192.168.10.2), a DNS server (192.168.10.3), and an email server (192.168.10.4). End-user devices were distributed across both switches: PC0 was connected directly to Switch0, while Laptop0 was connected to Switch1, extending network access.

The servers were then configured to provide their respective services. The DHCP server was set up with a pool to automatically assign IP addresses, subnet masks, and default gateway information to client devices like PC0 and Laptop0. The DNS server was populated with records to resolve hostnames to the correct IP addresses within the local network, primarily pointing to the web and email servers. The web server was configured with hosting software to deliver web pages and

content. Simultaneously, the email server was prepared to handle sending, receiving, and storing messages for network users.

Following configuration, network functionality was thoroughly validated. The successful operation of the DHCP service was confirmed when both PC0 and Laptop0 obtained their IP configurations automatically from the designated server. Furthermore, these clients were able to access the hosted website by its domain name, a process facilitated by the DNS server, and could successfully send and receive test emails through the local email server. This verified the seamless interoperability of all deployed network services

